Robust Multicolor Fiducials for Structure-From-Motion Pipelines

Purpose of markers
Structure-From-Motion (SfM) pipelines use distinctive local features to compute camera poses, and are challenged by featureless or repeating-detail surfaces. Fiducial markers (tags) can be placed on or around an object to provide high-contrast anchor points.

Problem space, Deformations, and Robustness
In practice these tags are affixed to surfaces that are moving and rotating, and are photographed from multiple angles, causing variable distortion and occlusion to the tags. In addition to geometric distortions, cameras may operate with a wide aperture to enable low ISO and high shutter speed (to minimize digital noise as well as motion blur) which results in a relatively shallow depth-of-field. Even with a lot of light, areas in the foreground and background will be slightly defocused, including tags located there. This also challenges the consistency and reliability of feature matching. A robust tag design should remain both visible and uniquely identifiable despite these deformations as well as those imposed by the feature detection algorithms.

April tags limitations
April tags, one commonly used black-and-white fiducial tag, are technically visually distinct from one another, but SfM's general-purpose detectors simply can't leverage that ID. a feature detector such as SURF or SIFT blurs the image to look for features like blobs, edges, and corners. After the regular visual deformations and this Gaussian blurring step, April tags below a certain size are not easily visually separable. Most dense black and white tags (ARTag, April tag, QR code, Stag, Maxicode, etc) devolve into a gray smudge when blurred, and lose all distinctness.

SIFTtag (smooth gradient dot)
SfM pipelines rely on a variety of feature detection and tracking algorithms, sometimes proprietary ones, to detect and track features across frames. Classical (SIFT, SURF, ORB, FAST, MSER, BRISK, or CNN) approaches locate features that are invariant to scale, rotation, and illumination. Techniques like SIFT apply multi-scale Gaussian blurring to the image and identify extrema in the Difference-of-Gaussians (DoG) space, to isolate distinct *blobs* that remain stable across transformations.

An optimal tag for SIFT feature detection has been developed and looks like a smooth gradient dot, characterized by its lack of high-frequency edge content:
https://www.researchgate.net/publication/220839283_Maximum_Detector_Response_Markers_for_SIFT_and_SURF

Unlike April tags, ARTtags, and others, SIFTtag exhibits minimal deformation under transformation. A smooth gradient dot has no high frequency edges, it remains almost unaffected by blurring, and is largely rotation, scale, and deformation invariant.

SIFTtag robustness vs. precision

While radial isotropy and a lack of sharp edges makes this tag maximally robust, it also makes it minimally precise for pose estimation (due to the flat gradient slope in the center of the dot and at the edges). It is desirable to combine the smooth gradient dots with some sort of high contrast sharp-edged features, so that the detector has enough information to *robustly* and also *precisely* estimate the camera poses.

Entropy vs. Robustness

Fine features such as patterns of dots or lines provide a better 'descriptor' but vanish entirely under defocus or motion. Large blocks of alternating brightness provide sharp edges that also persist through blurring. By surrounding a smooth gradient dot with these blocks, we could make one composite tag which contains both low-frequency robustness and high-frequency precision.

Self-Similarity

Self-similar tags, especially when arranged in a repeating pattern, increase the risk of false positive matches in SfM pipelines. Identical tags produce nearly indistinguishable descriptors (e.g. SIFT, ORB), making it hard to tell which tag is which across frames. Visually separable tags reduce ambiguity. The regularity of the pattern-of-tags is made irrelevant if the tags themselves are visually distinct, as the matching algorithm can then more easily reject false positives.

'Poly Cues' Polychromatic Polygons

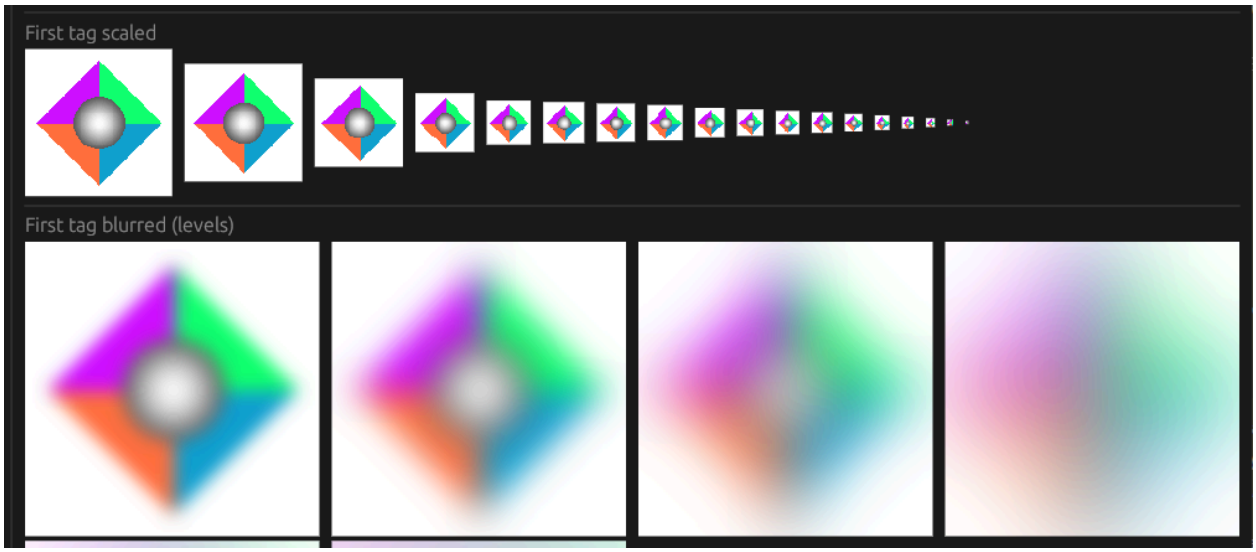
Conventionally, fiducial tags are black and white to maximize contrast. Since we use a color camera, one easy way to make otherwise identical tags visually unique is to simply make them different colors. Polycues are generated using a set of unique colors picked from a colorspace to maximize ΔE (perceptual color difference metric). These tags consist of a regular polygon, divided into slices radially. Each slice is a different color. In the center is a smooth Gaussian dot. This design contains the benefits of a SIFTtag, plus rotational asymmetry, tag to tag differentiability, and sharp edges and junctions for precision pose estimation.

Number of sides

The choice of sides has two confounding constraints. A smaller number of divisions means each is larger, and will more easily survive the DoG and camera defocus.
But also, a small number of sides, 3 for instance, causes the vertices to be very pointy and therefore extra susceptible to defocus, motion blur, and DoG. 4-gons have square corners. Corners, edges, and 'blobs' are all detected as trackable features by SfM software. As you add

sides the corners get closer and closer to 180 degrees, and the N-gon approximates a circle, lacking trackable corners. The program can only generate so many unique tags based on its method of choosing unique colors from a color space to maximize their difference. It can generate up to 62 3-sided tags, 46 4-sided tags, 37 5-sided tags, 31 6-sided tags, etc.

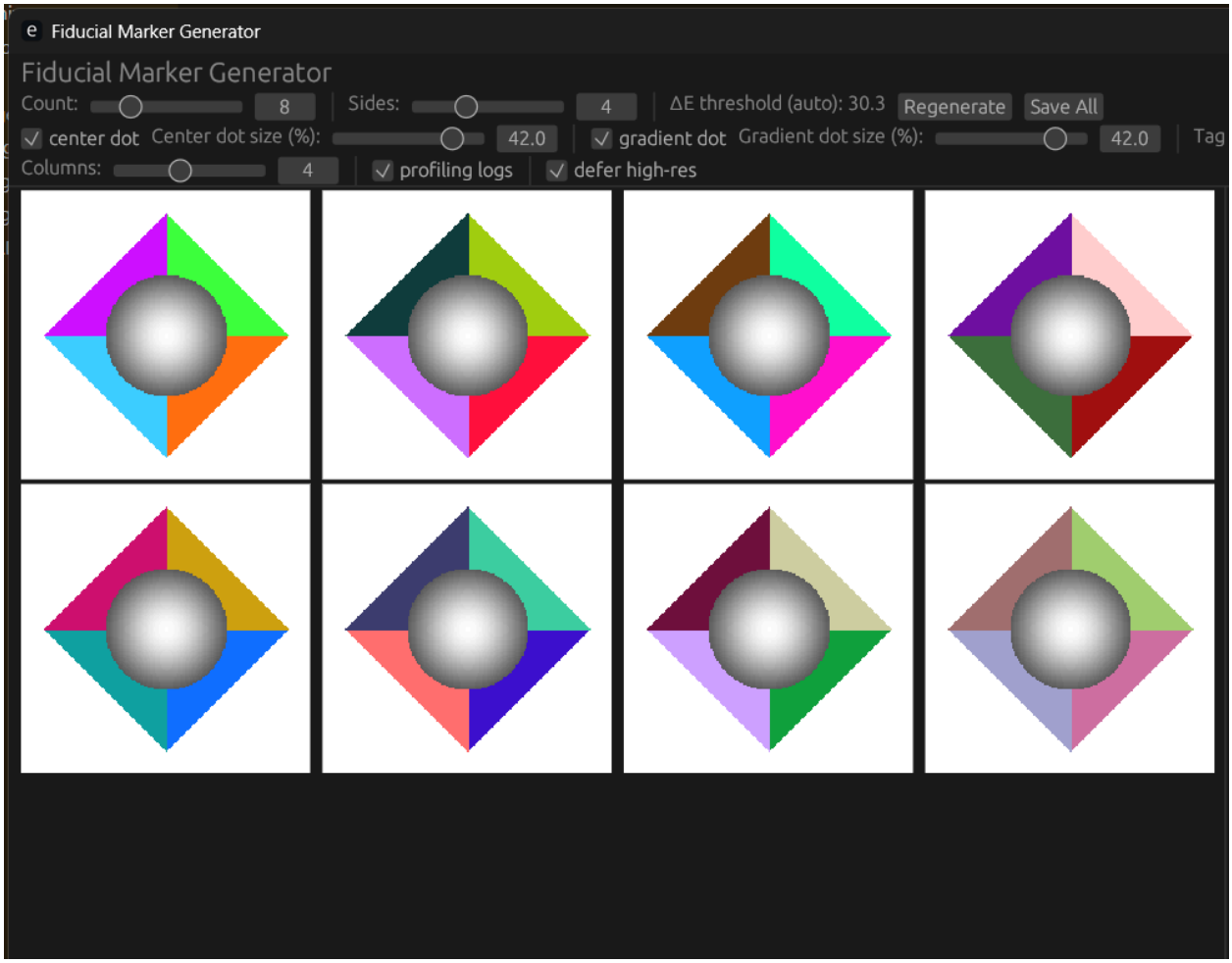
Seemingly 4 sides are best to balance all constraints, unless a small number of very large tags are to be used, in which case more than 4 sides may be beneficial.



Poly Cues are uniquely identifiable even when very low resolution or when blurred or defocused.

	Differentiable	Blur - Robust	Asymmetric
Dots / Grids	X	X	X
SIFT tags	X	✓	X
April tags	✓	X	✓
Poly Cues	✓	✓	✓

Monochrome (half-size)
First tag scales
First tag blurred (levels)


Fiducial Marker Generator
Fiducial Marker Generator
Count:
8
Sides:
4
ΔE threshold (auto): 30.3
Regenerate
Save All
center dot
Center dot size (%):
42.0
gradient dot
Gradient dot size (%):
42.0
Tag
Columns:
4
profiling logs
defer high-res